FIGURE 1 · METHODOLOGY

The SHARP–AEGIS pipeline

SHARP measures the smallest honest model; AEGIS governs whether it is fit for its job; a human signs every gate. The machine only recommends.

1 DATA INTAKE & SCOPING

"What is the target, how many defaults, any protected attributes?"

Size by the scarcest resource: EPV = defaults ÷ features. Firm attrs → fairness N/A; person attrs → fairness live.

2 SHARP — THE FEATURE FUNNEL

Each stage removes a different kind of lie. Pre-registered, fixed across all datasets.

1. **Screen** — drop obvious noise and leakage
2. **Correlation grouping** — collapse look-alike features into one "player"
3. **Shapley ranking** — credit each player fairly over every team it could join
4. **Shadow-null** — keep only features that beat an injected random decoy
5. **Parsimony** — smallest set with DeLong p ≥ 0.05 and rel-Gini loss ≤ 5%

◆ **columns lock → THE model is fit here (WOE-logistic on the final columns)**

3 ALWAYS-ON SCORING HYGIENE

Isotonic calibration
"we said 10%" ≈ 10-in-100 default

PD floor 0.03%
no zero-risk borrower (Basel CRE 32.3)

4 AEGIS — THE GATE LADDER

Machine recommends + evidence → human signs → hash-chained ledger. Gates decide governance, not analytics.

G0	**Eligibility & scope**	what may we model? protected attrs & materiality tier
G1	**Data sufficiency**	is there enough — EPV — to proceed honestly?
G2	**Feature governance**	features legitimate? no proxies, no leakage
G3	**Model & method**	confirm the pre-registered method — do not retune
G4	**Performance & calibration backtest**	per-grade Wilson CIs, Hosmer–Lemeshow, ECE (held-out 1/5)
G5	**Deliverables**	decisioning note + 6-case pack; no cutoff/pricing invented
GF	**Final verdict**	are the weaknesses tolerable at this model's tier?

sealed by a hash chain (genesis → G_n → …) so the record cannot be edited after signing

continued →

FIGURE 1 · CONTINUED

The governed verdict — and the nine runs

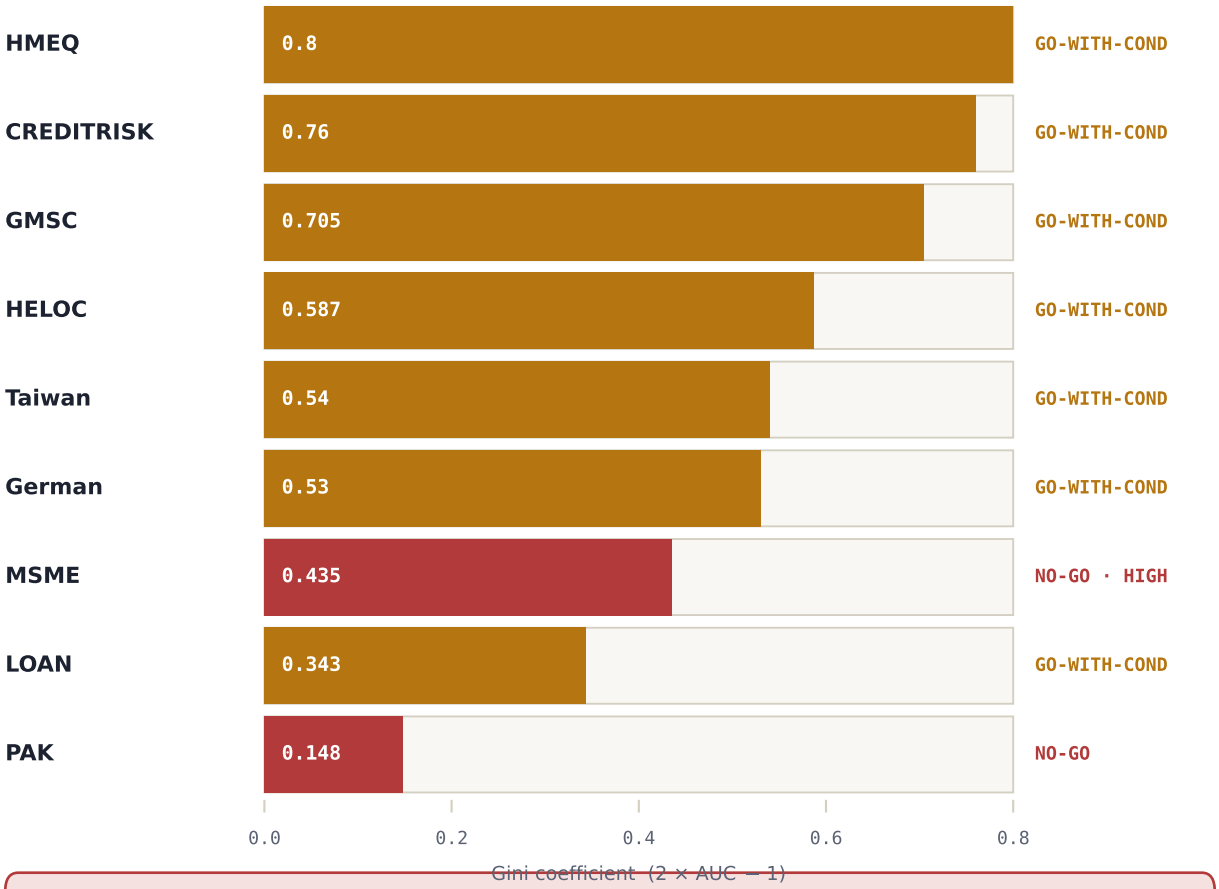
5 THE GOVERNED VERDICT

Read off fitness-for-tier — strictness scales with how much the model matters (LOW → MEDIUM → HIGH).

GO	GO-WITH-CONDITIONS	NO-GO
fit for its tier as-is	usable with named monitoring / limits	refused — by judgment or by rule

6 NINE RUNS — AND WHY GINI DOES NOT DECIDE

Each model's Gini (rank-ordering power) vs its verdict. No single horizontal line separates the colours.



The crossover

MSME (Gini 0.435) is a NO-GO while LOAN (Gini 0.343, lower discrimination) is GO-WITH-CONDITIONS. MSME failed on EPV ≈ 1.2 and HIGH-tier bars — not on Gini. The GO/NO-GO decision is multi-factor by design, not a single accuracy cutoff.

Figure 1. The SHARP–AEGIS governance pipeline. SHARP selects the smallest defensible feature set and reports discrimination with a confidence interval; AEGIS governs deployment through a human-signed, hash-chained gate ladder. Across nine credit datasets the same process produced seven GO-WITH-CONDITIONS and two NO-GO verdicts — one by human judgment (PAK: redlining proxies), one by rule (MSME: HIGH-tier data-sufficiency bars) — demonstrating risk-proportionate rigor, not one cutoff.

Gini values are per-run point estimates from the recorded runs; grade-band edges require an external/stated internal scale citation before publication.